TWO HIDDEN LAYERS FOR MAX: A CATERPILLAR CONSTRUCTION

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ABSTRACT

Every continuous piecewise affine function is a finite signed sum of maxima of affine functions, making maximum representation a central problem for ReLU networks. We give a nonrecursive caterpillar construction for the maximum of N inputs with two hidden layers of widths $O(N^2)$ and $2^{O(N \log N)}$. The construction has a direct geometric interpretation, and its correctness is proved in Lean by induction.[1]

1 INTRODUCTION

In this paper we investigate the representation capabilities of ReLU networks: What functions are representable by such networks of different depth and widths? As ReLU neural networks represent continuous piecewise-linear (CPWL) functions, a sharp question is: given any CPWL function, what is the ReLU network of smallest depth or smallest width representing this function? The maximum is a universal building block for such CPWL functions. Write

$$\max_N(x) := \max_{0 \le i < N} x_i. \tag{1}$$

Theorem (Wang and Sun, 2005)**.** *Let $d \ge 1$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous and affine on each member of a finite polyhedral cover. There are $R \in \mathbb{N}$, affine maps and signs*

$$A_t : \mathbb{R}^d \longrightarrow \mathbb{R}^{d+1}, \qquad \varepsilon_t \in \{-1, 1\} \qquad (1 \le t \le R) \tag{2}$$

such that

$$f(x) = \sum_{t=1}^{R} \varepsilon_t \max_{d+1}(A_t(x)) \qquad \textit{for every } x \in \mathbb{R}^d. \tag{3}$$

Thus a two-hidden-layer representation of $\max_{d+1}$ gives one for every CPWL function on $\mathbb{R}^d$: precompose with the affine maps and add the networks in parallel.

This reduction made the geometry of the maximum a central approach to the representation problem well before the work of Brunck et al. (2026). Hertrich et al. (2023) studied depth through Newton polytopes, using that the maximum is the support function of a simplex, while Grillo et al. (2025) obtained depth lower bounds under compatibility with the braid fan. Brunck et al. (2026) then constructed two-hidden-layer ReLU representations for arbitrary CPWL functions in every dimension.

Theorem 1 (Brunck et al., 2026)**.** *Every continuous piecewise affine function $f : \mathbb{R}^d \to \mathbb{R}$ with finitely many affine pieces can be represented exactly by a ReLU network with two hidden layers.*

Our contribution is to specialize their general construction to $\max_N$ and simplify it. Taking $2^{\Theta(N^2)}$ as the second-layer width baseline via applying the Brunck et al. (2026) recursive construction, we pursue two aims:

1. Give a closed, nonrecursive construction for $\max_N$ with a direct geometric interpretation.
2. Reduce its second hidden-layer width.

[1] Our results were obtained through proof search in Lean using Claude Code and Codex. The Lean code is available in the accompanying repository https://github.com/JayPiZimmermann/Max_Caterpillar. Links to the Lean formalization appear in theorem titles and equation numbers; equation numbers are clickable.

Write $\sigma(t) := \max(0, t)$ applied coordinatewise to vectors. A network of widths (h_1, h_2) has parameters

$$\begin{aligned} W_1 &\in \mathbb{R}^{h_1 \times N}, \quad b_1 \in \mathbb{R}^{h_1}, \\ W_2 &\in \mathbb{R}^{h_2 \times h_1}, \quad b_2 \in \mathbb{R}^{h_2}, \\ w_3 &\in \mathbb{R}^{h_2}, \qquad b_3 \in \mathbb{R}, \end{aligned} \tag{4}$$

and evaluates as the function

$$x \longmapsto w_3^\top \sigma\big(W_2 \sigma(W_1 x + b_1) + b_2\big) + b_3. \tag{5}$$

The output is affine, there are no skip connections, and equality is required on all of $\mathbb{R}^N$.

Theorem 2 (`Lean_1`, `Lean_2`, proof)**.** *For every $N \geq 3$, define*

$$C_N = \sum_{m=2}^{N-1} \binom{N}{m} \frac{m!}{2} (N - m) m^{N-m-1}. \tag{6}$$

There is a two-hidden-layer ReLU network computing $\max_N$ with widths

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = 3^{N-2} C_N - N(N-1) + 2. \tag{7}$$

In particular, these widths satisfy, as $N \to \infty$,

$$h_1 = O(N^2), \qquad h_2 = 2^{O(N \log N)}. \tag{8}$$

The network is specified by the finite construction below. Widths are upper bounds, realized exactly by zero padding when necessary.

2 THE FINITE CONSTRUCTION

Two one-line identities drive the construction. For reals s, t and every linear function ℓ on $\mathbb{R}^N$,

$$\max(s, t) = t + \sigma(s - t), \tag{9}$$

$$\sigma(\ell(x)) = \sigma(-\ell(x)) + \ell(x). \tag{10}$$

The first suggests computing the maximum along a path with attached leaves. Alternating sums of support functions of caterpillar polytopes give the correct support-function class; support functions of their proper faces, together with linear functions, correct the remaining error to pointwise equality. The second reverses an inner hinge at the cost of a linear term; it merges equal units in Section 4.

Fix $N \geq 3$ and write $x = (x_0, \ldots, x_{N-1}) \in \mathbb{R}^N$.

Definition 3 (Labelled caterpillar, `Lean_3`)**.** A *caterpillar description* is a permutation–set pair

$$\pi \in \mathrm{Sym}\{0, \ldots, N-1\}, \qquad \varnothing \neq S \subseteq \{1, \ldots, r\}, \qquad r := N - 2. \tag{11}$$

Position j carries the label $\pi(j+1)$, for $0 \leq j \leq r$, and the label $\pi(0)$ is the *apex*. The *spine positions* are $\{0\} \cup S$. We also call S the *cut set*: a cut at $j \in S$ separates positions $j-1$ and j in the ordered list. The last spine position, and the *parent* of a position $j \geq 1$, the nearest spine position before it, are

$$q := \max S, \qquad p(j) := \max\big(\{0\} \cup \{i \in S : i < j\}\big) \qquad (1 \leq j \leq r). \tag{12}$$

Joining each position $j \geq 1$ to its parent gives a directed tree on the labels $\pi(1), \ldots, \pi(N-1)$, the *labelled caterpillar*: its spine is the path through the spine positions, and every other label is a leaf attached to the preceding spine vertex (Figure 1). The apex is separate from this tree.

Definition 4 (Caterpillar function and face functions, `Lean_4`, `Lean_5`)**.** For labels i, j, for the tree edges, and for the head, put

$$\ell_{ij}(x) := x_i - x_j, \tag{13}$$

$$L_{\pi,S} := \big\{\ell_{\pi(p(j)+1),\pi(j+1)} : 1 \leq j \leq r\big\}, \tag{14}$$

$$h_{\pi,S} := \ell_{\pi(q+1),\pi(0)}. \tag{15}$$

The *edge functions* $L_{\pi,S}$ subtract each child's coordinate from its parent's, one for each of the r tree edges, and the *head function* $h_{\pi,S}$ compares the last spine vertex with the apex. For a linear function h, a finite set L of linear functions, and a *state* $\tau(\ell) \in \{0,\ \ell,\ \sigma \circ \ell\}$ chosen for every $\ell \in L$, define

$$U^{\tau}_{h,L}(x) := \sigma\Big(h(x) + \sum_{\ell \in L} \tau(\ell)(x)\Big). \tag{16}$$

Thus each $\ell \in L$ is omitted, kept linearly, or kept as a hinge. Write $U_{h,L} := U^{\tau}_{h,L}$ for the all-hinge state $\tau(\ell) = \sigma \circ \ell$. The *caterpillar function* of (π, S) and its *face functions* are

$$U_{\pi,S} := U_{h_{\pi,S},L_{\pi,S}}, \qquad U^{\tau}_{\pi,S} := U^{\tau}_{h_{\pi,S},L_{\pi,S}}. \tag{17}$$

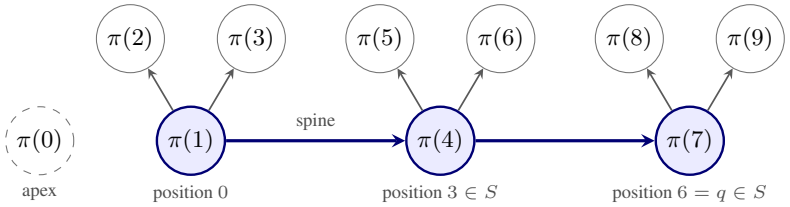


Figure 1: A labelled caterpillar with $N = 10$ and $S = \{3, 6\}$. The spine $\pi(1) \to \pi(4) \to \pi(7)$ has two leaves at each vertex; all arrows point from parent to child. The apex $\pi(0)$ is separate from the tree. The last spine position is $q = 6$, so the head function is $h_{\pi,S}(x) = x_{\pi(7)} - x_{\pi(0)}$.

The term *face function* comes from the following geometry. Identify linear functions with their coefficient vectors. For a polytope P, its *support function*, and the polytope whose support function is $U_{h,L}$, are

$$h_P(x) := \max_{p \in P} \langle p, x \rangle, \tag{18}$$

$$Q_{h,L} := \mathrm{conv}\Big(\{0\} \cup \Big(h + \sum_{\ell \in L} [0, \ell]\Big)\Big), \qquad U_{h,L} = h_{Q_{h,L}}. \tag{19}$$

Indeed, the segment $[0, \ell]$ has support function $\sigma \circ \ell$, Minkowski sums add support functions, and adjoining 0 applies the outer ReLU. For a caterpillar the edge vectors are independent, so $Q_{h,L}$ is a pyramid over a parallelotope. A face of $Q_{h,L}$ containing the apex 0 and another point is the convex hull of 0 and a face of the base. Since the base is a translated sum of segments, the face exposed by a linear functional is obtained by maximizing on each segment separately, which selects the endpoint 0, the endpoint ℓ, or the whole segment. These have support functions 0, ℓ, and $\sigma \circ \ell$, and the outer ReLU adjoins the apex. Hence the face functions (16) include the support function of every face of $Q_{h,L}$ through the apex.

Definition 5 (Dictionary, `Lean_6`)**.** The *dictionary* $\mathcal{D}$ is the finite set of functions $\mathbb{R}^N \to \mathbb{R}$ consisting of every face function $U^{\tau}_{\pi,S}$, over all caterpillar descriptions (π, S) and all states τ, together with the *auxiliary functions*

$$\begin{aligned} &x \mapsto \sigma(x_i), \quad x \mapsto \sigma(-x_i) && (0 \le i < N), \\ &x \mapsto \sigma(x_i - x_j) && (0 \le i < j < N). \end{aligned} \tag{20}$$

Heads and edge functions are coordinate differences, so face functions and pair hinges are translation invariant: with $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^N$,

$$U^{\tau}_{\pi,S}(x + t\mathbf{1}) = U^{\tau}_{\pi,S}(x) \qquad (t \in \mathbb{R}). \tag{21}$$

Only the coordinate features $\sigma(\pm x_i)$ change under $x \mapsto x + t\mathbf{1}$. Theorem 7 below proves $\max_N \in \mathrm{span}\,\mathcal{D}$; the following rule then fixes one real coefficient for each dictionary function.

An exact coefficient rule (`Lean_7`) Let

$$\mathcal{B} := \{S \subseteq \mathcal{D} : S \text{ is a basis of } \operatorname{span}\mathcal{D}\}. \tag{22}$$

This set is nonempty because the finite spanning set $\mathcal{D}$ contains a basis. For $S \in \mathcal{B}$, form

$$G_S := \big(\langle g, h\rangle\big)_{g,h\in S}, \tag{23}$$

$$b_S := \big(\langle g, \max_N\rangle\big)_{g\in S}. \tag{24}$$

using the inner product

$$\langle f, g\rangle := \int_{[-1,1]^N} f(x)g(x)\,dx. \tag{25}$$

As G_S is positive definite it is invertible; let us denote its inverse matrix by G_S^{-1}. For each $g \in \mathcal{D}$, average over the bases containing g:

$$c_g := \frac{1}{|\mathcal{B}|}\sum_{\substack{S\in\mathcal{B}\\ g\in S}} (G_S^{-1}b_S)_g. \tag{26}$$

Lemma 6 (Canonical expansion, `Lean_8`)**.** *If* (40) *holds, the coefficients* (26) *satisfy*

$$\max_N = \sum_{g\in\mathcal{D}} c_g g. \tag{27}$$

Proof of Lemma 6. Fix $S \in \mathcal{B}$. By (40), there is $d \in \mathbb{R}^S$ such that

$$\max_N = \sum_{g\in S} d_g g. \tag{28}$$

Taking the inner product with each $h \in S$ gives

$$\begin{aligned}(b_S)_h &= \sum_{g\in S}\langle h, g\rangle d_g = (G_S d)_h,\\ d &= G_S^{-1}b_S.\end{aligned} \tag{29}$$

Interchanging the finite sums in (26) therefore yields

$$\sum_{g\in\mathcal{D}} c_g g = \frac{1}{|\mathcal{B}|}\sum_{S\in\mathcal{B}}\sum_{g\in S}(G_S^{-1}b_S)_g\, g = \frac{1}{|\mathcal{B}|}\sum_{S\in\mathcal{B}}\max_N = \max_N. \tag{30}$$

This finishes the proof of Lemma 6. □

3 WHY THE DICTIONARY SPANS THE MAXIMUM

The maximum $\max_N$ is the support function of $\operatorname{conv}\{e_0, \dots, e_{N-1}\}$. To work with a full dimensional polytope, let us do the following substitution: for

$$\Delta := \operatorname{conv}\{0, e_1, \dots, e_{N-1}\} \subseteq H_0 := \{x \in \mathbb{R}^N : x_0 = 0\} \tag{31}$$

we observe for an arbitrary input $x \in \mathbb{R}^N$ that

$$\max_N(x) = x_0 + h_\Delta(x - x_0\mathbf{1}). \tag{32}$$

To prove that $\max_N \in \operatorname{span}\mathcal{D}$ it suffices to show that h_Δ lies in the span of the restrictions of $\mathcal{D}$ to H_0. Indeed, $x_0 = \sigma(x_0) - \sigma(-x_0)$ lies in the span of the auxiliary functions (20), and by (21) the substitution $x \mapsto x - x_0\mathbf{1}$ fixes every dictionary function except the coordinate features $\sigma(\pm x_i)$, which become $\sigma(\pm(x_i - x_0))$; by (10) these are linear combinations of auxiliary functions. Hence $x \mapsto g(x - x_0\mathbf{1})$ belongs to $\operatorname{span}\mathcal{D}$ for every $g \in \mathcal{D}$, and (32) gives the claimed implication.

All polytopes in the geometric argument below lie in the $(N-1)$-dimensional space H_0. We restrict heads and edge functions to H_0 and identify them with their coefficient vectors there. Let $\mathcal{L}_k$ denote the real span of support functions of polytopes in H_0 of affine dimension at most k. In particular, $\mathcal{L}_0$ is the space of linear functions, since the support function of a point p is $x \mapsto \langle p, x\rangle$.

For a nonzero vector

$$u \in H_0 \setminus \{0\}, \tag{33}$$

the exposed face of a polytope $P \subseteq H_0$ with normal u is the set of points of P maximizing the linear function $p \mapsto \langle p, u\rangle$:

$$P^u := \{p \in P : \langle p, u\rangle = h_P(u)\}. \tag{34}$$

Geometrically, it is where the supporting hyperplane with outward normal u meets P. The opposite face P^{-u} consists of the minimizers. Define the function

$$D_u h_P(x) := h_{P^u}(x) - h_{P^{-u}}(x) \qquad (x \in H_0). \tag{35}$$

Here u selects the two faces, while x is the input to their support functions.

For a finite linear combination g of support functions, define its one-sided directional derivative at u by

$$\begin{aligned} \partial_u g(x) &:= \lim_{t \downarrow 0} \frac{g(u + tx) - g(u)}{t}, \\ \partial_u h_P = h_{P^u}, &\qquad D_u g = \partial_u g - \partial_{-u} g. \end{aligned} \tag{36}$$

Thus the map $h_P \mapsto h_{P^u}$ extends linearly to linear combinations of support functions, independently of their decomposition. If $\dim P \le k$, either $\langle u, \cdot\rangle$ is constant on P, so $D_u h_P = 0$, or both selected faces have dimension at most $k - 1$. Consequently,

$$D_u \mathcal{L}_k \subseteq \mathcal{L}_{k-1} \qquad (k \ge 1). \tag{37}$$

The signed caterpillar sum. We seek a combination that agrees with h_Δ up to a function in $\mathcal{L}_{N-2}$. Adding $x_{\pi(0)}$ makes the caterpillar function the following maximum

$$x_{\pi(0)} + U_{\pi,S}(x) = \max\left\{ x_{\pi(0)},\, x_{\pi(q+1)} + \sum_{\ell \in L_{\pi,S}} \sigma(\ell(x)) \right\}. \tag{38}$$

Averaging over permutations gives the required symmetry

$$F_N(x) := \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \ne S \subseteq \{1,\dots,r\}} (-1)^{N+|S|} \big(x_{\pi(0)} + U_{\pi,S}(x)\big), \tag{39}$$

where the signs $(-1)^{N+|S|}$ are chosen to cancel the extra contributions from support functions of exposed caterpillar faces, modulo $\mathcal{L}_{N-3}$. Establishing the required cancellations is a main step of the proof below.

Theorem 7 (Dictionary sufficiency, `Lean_9`, proof)**.** *For every $N \ge 3$, the dictionary of Definition 5 satisfies*

$$\max_N \in \operatorname{span} \mathcal{D}. \tag{40}$$

Proof of Theorem 7. Recall from this section's introduction that it suffices to show that h_Δ lies in the span of the restrictions of $\mathcal{D}$ to H_0.

Define the error by

$$\begin{aligned} f : H_0 &\longrightarrow \mathbb{R}, \\ f(x) &:= F_N(x) - h_\Delta(x) \\ &= F_N(0, x_1, \dots, x_{N-1}) - \max\{0, x_1, \dots, x_{N-1}\}. \end{aligned} \tag{41}$$

Step 7.1 reduces dictionary membership to $f \in \mathcal{L}_{N-2}$. Step 7.2 shows that it suffices to prove $D_u f \in \mathcal{L}_{N-3}$ for every direction u. Step 7.3 proves these bounds by induction on N: dimension bounds and cancellation handle all directions except those with one coordinate different from all the others, which share a common value. In that case, deleting the exceptional coordinate invokes the induction hypothesis.

Step 7.1 (Reduce to a lower-dimensional error). In this step we reduce the problem to proving

$$f = F_N - h_\Delta \in \mathcal{L}_{N-2}. \tag{42}$$

Assume this relation temporarily, and write $Q_{\pi,S} := Q_{h_{\pi,S},L_{\pi,S}}$, so that $U_{\pi,S} = h_{Q_{\pi,S}}$ by (19). Since the apex terms in (39) are linear, the assumed relation gives

$$\frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}h_{Q_{\pi,S}} - h_\Delta \in \mathcal{L}_{N-2}. \tag{43}$$

We will correct this error by differentiating its support-function terms via ∂_u to lower the residual dimension until the residual is linear. The resulting support functions of proper faces of the $Q_{\pi,S}$ already lie in the dictionary span: the support function of the convex hull of the apex with a base face is a face function from Definition 4, given by (16), while the support function of a base face is a linear function plus inner hinges, represented by (20). The apex alone has zero support function, and $h_{Q_{\pi,S}} = U_{\pi,S}$ is itself a dictionary function. Differentiating the simplex support-function term also produces support functions of proper simplex faces. Their membership in the dictionary span is supplied by induction on the number of vertices.

Claim (induction on $|T|$). For every nonempty $T \subseteq \{0, e_1, \dots, e_{N-1}\}$, the support function of $\Delta_T := \operatorname{conv} T$ lies in the dictionary span.

Start ($|T| = 1$). The support function of a singleton is linear and therefore lies in the dictionary span.

Step ($|T| \geq 2$). Fix T and assume that $h_{\Delta_A} \in \operatorname{span}\mathcal{D}$ for every nonempty vertex set A with $|A| < |T|$, in particular for every nonempty proper subset of T. Consider a current vertex set $A \subseteq \{0, e_1, \dots, e_{N-1}\}$ with $|A| \geq 2$. To remove $v \in A$, choose a direction whose value at v is strictly larger than its common value at all other vertices. The positive exposed face is then the singleton $\{v\}$, and the negative exposed face is the simplex on the remaining vertices. Writing $\Delta_A := \operatorname{conv} A$, we have

$$\begin{aligned} u_v &:= \begin{cases} -\sum_{i=1}^{N-1} e_i & v = 0, \\ v & v \neq 0, \end{cases} \\ (\Delta_A)^{u_v} &= \{v\}, \qquad (\Delta_A)^{-u_v} = \Delta_{A\setminus\{v\}}, \\ D_{u_v} h_{\Delta_A}(x) &= \langle v, x\rangle - h_{\Delta_{A\setminus\{v\}}}(x). \end{aligned} \tag{44}$$

Choose an ordering $v_1, \dots, v_{N-|T|}$ of the vertices outside T. Denote the error associated with the remaining vertex set T by

$$\varepsilon_T := D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} \left(\frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}h_{Q_{\pi,S}} - h_\Delta \right). \tag{45}$$

The operators act from right to left, removing v_1, then v_2, and so on. In (44), the support function of the remaining simplex carries a minus sign, while the removed singleton contributes the linear function $x \mapsto \langle v, x\rangle$. Every subsequent D_u annihilates linear functions. Thus the $N - |T|$ deletions give

$$D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} h_\Delta = (-1)^{N-|T|} h_{\Delta_T} + \ell_T, \tag{46}$$

where ℓ_T is linear. The original error contains $-h_\Delta$, so the coefficient of h_{Δ_T} in ε_T is $-(-1)^{N-|T|} = (-1)^{N-|T|+1}$. All other terms are linear functions or support functions of faces of the $Q_{\pi,S}$, since each D_u replaces a support function by a difference of support functions of exposed faces. The coefficient of h_{Δ_T} in ε_T is ± 1, and all other terms already lie in the dictionary span. Thus proving $\varepsilon_T \in \operatorname{span}\mathcal{D}$ completes the induction step. To prove this membership, we use the additional information $\varepsilon_T \in \mathcal{L}_{|T|-2}$, obtained by repeated application of (37). This bound starts the inner induction below: each reduction lowers the dimension bound by one while subtracting a function in the dictionary span, until the remaining error lies in $\mathcal{L}_0$ and is therefore linear. For the

full vertex set, the operator composition is empty and means the identity, so ε_T is exactly the error in (43).

For this fixed T, define the remaining errors as functions $\varepsilon_T^{(j)} : H_0 \to \mathbb{R}$ recursively by

$$\begin{aligned} \varepsilon_T^{(0)}(x) &:= \varepsilon_T(x), \\ \varepsilon_T^{(j+1)}(x) &:= \varepsilon_T^{(j)}(x) - \sum_{u \in E_{T,j}} \big(\partial_u \varepsilon_T^{(j)}\big)(x). \end{aligned} \tag{47}$$

Here $x \in H_0$ and $0 \le j < |T| - 2$. At stage j, the finite set of directions $E_{T,j} \subseteq H_0$ is chosen from the current error by the construction below. Thus $\varepsilon_T^{(j)}$ is the error left after j derivative sums have been subtracted from ε_T; the superscript counts stages and is not a power.

Claim (induction on the reduction stage j). The sets $E_{T,j}$ can be chosen successively so that, for $0 \le j \le |T| - 2$,

$$\begin{aligned} \varepsilon_T^{(j)} &\in \mathcal{L}_{|T|-2-j}, \\ \varepsilon_T - \varepsilon_T^{(j)} &\in \operatorname{span} \mathcal{D}. \end{aligned} \tag{48}$$

Each remaining error is a linear combination of h_{Δ_T}, support functions of proper faces of Δ_T, support functions of faces of the $Q_{\pi,S}$, and linear functions. The second condition says that the part already removed is represented by the dictionary.

Start ($j = 0$). By definition, $\varepsilon_T^{(0)} = \varepsilon_T$. Its dimension bound and support-function expression were established above, and the removed part is zero.

Step ($j < |T| - 2$). Assume the directions for the first j stages have been chosen and the claim holds at stage j. Put $k := |T| - 2 - j \ge 1$. We require each new direction u to be nonorthogonal to the simplex direction space $\operatorname{dir} \Delta_T := \operatorname{span}\{v - w : v, w \in T\}$. Its values on T are then not all equal, so the maximizing vertices form a nonempty proper subset:

$$\begin{aligned} A_u &:= \{v \in T : \langle u, v \rangle = h_{\Delta_T}(u)\}, \qquad \varnothing \ne A_u \subsetneq T, \\ \partial_u h_{\Delta_T} &= h_{\Delta_{A_u}} \in \operatorname{span} \mathcal{D}. \end{aligned} \tag{49}$$

The last membership uses the outer induction hypothesis, since $|A_u| < |T|$. An orthogonal direction would leave h_{Δ_T} unchanged and would not allow this use of the hypothesis.

Choose an auxiliary expression for $\varepsilon_T^{(j)}$ by support functions of polytopes of dimension at most k, and collect their distinct k-dimensional direction spaces. For each such space V, choose $u_V \in V^\perp$ not orthogonal to $\operatorname{dir} \Delta_T$ or to any other collected space V'. This is possible because $\dim \Delta_T = |T| - 1 > k$: each of $\operatorname{dir} \Delta_T$ and V' contains a vector outside V, whose nonzero projection onto $V^\perp$ is not orthogonal to that vector. Adding sufficiently small multiples of these projections avoids each orthogonality condition in turn while preserving those already avoided. Choose $E_{T,j} := \{u_V : V \text{ is a collected direction space}\}$, with $E_{T,j} = \varnothing$ if there are none. In the sum of derivatives over $E_{T,j}$, each term coming from a k-dimensional polytope is preserved exactly once. Every other derivative of that term is the support function of a proper face, while terms of lower dimension remain in $\mathcal{L}_{k-1}$. Subtracting this derivative sum therefore cancels all k-dimensional terms. By the recursive definition (47), the resulting error satisfies $\varepsilon_T^{(j+1)} \in \mathcal{L}_{k-1} = \mathcal{L}_{|T|-3-j}$.

Now compute these same derivatives using the support-function expression in the claim. The derivative of h_{Δ_T} lies in the dictionary span by (49). Differentiating the support function of an already proper simplex face still gives a support function of a proper face of Δ_T, so the same induction hypothesis applies. Differentiating the support function of a face of $Q_{\pi,S}$ gives a support function of a face of the same polytope, covered by the opening dictionary description, and linear functions remain linear. Thus the derivative sum lies in the dictionary span. By (47), $\varepsilon_T - \varepsilon_T^{(j+1)}$ is the sum of the previously removed part and this derivative sum, so it also lies in the dictionary span. Subtracting the derivative sum preserves the stated types of support-function terms. This proves the claim at stage $j + 1$.

At stage $j = |T| - 2$, the remaining error lies in $\mathcal{L}_0$ and is linear. Together with (48), this gives $\varepsilon_T \in \operatorname{span} \mathcal{D}$. Its vertex-deletion expression has coefficient $(-1)^{N-|T|+1} \in \{-1, 1\}$ on h_{Δ_T} and all other terms in the dictionary span, so $h_{\Delta_T} \in \operatorname{span} \mathcal{D}$. This completes the induction step on $|T|$. Taking T to be the full vertex set proves $h_\Delta \in \operatorname{span} \mathcal{D}$ under the temporary assumption (42). This finishes the proof of Step 7.1.

Step 7.2 (Why the facet bounds imply the class identity). For $N \geq 4$ and $f = F_N - h_\Delta$, we show that $f \in \mathcal{L}_{N-2}$ follows from

$$D_u f \in \mathcal{L}_{N-3} \qquad \text{for every } u \in H_0. \tag{50}$$

The idea is to express f modulo $\mathcal{L}_{N-2}$ using support functions of full-dimensional simplices. The facet bounds then force their weighted contributions to cancel modulo $\mathcal{L}_{N-2}$.

The base of each $Q_{\pi,S}$ is an affine cube. Order its edge vectors by a bijection $\omega : \{1, \dots, N-2\} \to L_{\pi,S}$, add them one at a time, and adjoin the apex:

$$K_{\pi,S,\omega} := \operatorname{conv}\left(\{0\} \cup \left\{ h_{\pi,S} + \sum_{a=1}^{k} \omega(a) : 0 \leq k \leq N-2 \right\}\right). \tag{51}$$

These $(N-2)!$ simplices form the *Kuhn subdivision*: in cube coordinates, each occupies one ordering region $1 \geq t_{\omega(1)} \geq \cdots \geq t_{\omega(N-2)} \geq 0$. For a hyperplane cut $P = P_+ \cup P_-$, the support-function identity

$$h_P + h_{P_+ \cap P_-} = h_{P_+} + h_{P_-} \tag{52}$$

shows that subdivision changes the sum of support functions only by support functions of the cutting sections. Hence

$$h_{Q_{\pi,S}} - \sum_{\omega} h_{K_{\pi,S,\omega}} \in \mathcal{L}_{N-2}. \tag{53}$$

Substituting (53) into (39) expresses f as a finite linear combination of support functions of $(N-1)$-dimensional simplices in H_0, plus a remainder in $\mathcal{L}_{N-2}$. The simplex Δ contributes its support function with coefficient -1, and the linear apex terms are included in the remainder.

For a simplex P, an outward facet normal ν_F exposes the facet F on one side and its opposite vertex v_F on the other. Thus

$$D_{\nu_F} h_P(x) = h_F(x) - \langle v_F, x \rangle. \tag{54}$$

Directions exposing no facet on either side give support functions of faces of dimension at most $N-3$. Consequently, (50) says that each weighted, signed sum of parallel-facet support functions belongs to $\mathcal{L}_{N-3}$.

Our assumption controls the weighted facet contributions, whereas our conclusion concerns f. To recover the original sum modulo $\mathcal{L}_{N-2}$, we use the same linear operator for every simplex, so cancellations between facet contributions survive reconstruction. Define

$$(\mathcal{T} g)(x) := \begin{cases} g(x) & x_1 \geq 0, \\ g(x - x_1 e_1) & x_1 < 0. \end{cases} \tag{55}$$

It satisfies

$$\mathcal{T} \mathcal{L}_k \subseteq \mathcal{L}_{k+1}. \tag{56}$$

Indeed, for a polytope P in $p_1 \geq 0$, $\mathcal{T} h_P$ is the support function of the convex hull of P and its projection onto $p_1 = 0$. This hull has dimension at most $\dim P + 1$. Translating an arbitrary polytope into that halfspace adds only a multiple of the segment support function $\sigma(x_1)$ as a correction.

Claim (induction on k). Every $(N-1)$-dimensional simplex $P \subseteq H_0$ with exactly k facets not parallel to e_1 satisfies

$$h_P - \mathcal{T}\left(\sum_{F \text{ facet of } P} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F \right) \in \mathcal{L}_{N-2}. \tag{57}$$

Here ν_F is an outward normal of F. Its sign against e_1 records whether the facet bounds lines parallel to e_1 from above or below; parallel facets contribute zero. Boundedness ensures both nonzero signs occur, so $k \geq 2$.

Start ($k = 2$). There is one upper facet F^+ and one lower facet F^-. Every line parallel to e_1 that meets P has its two intersection endpoints on these facets. Thus $h_P(x) = h_{F^+}(x)$ when $x_1 \geq 0$, and $h_P(x) = h_{F^-}(x)$ when $x_1 < 0$. When $x_1 \geq 0$, $\mathcal{T}$ evaluates at x, so subtracting $\mathcal{T}(h_{F^+} - h_{F^-})$ from h_P leaves h_{F^-}. When $x_1 < 0$, $\mathcal{T}$ evaluates both facet support functions at $x - x_1 e_1$, where they agree because the facets have the same projection. Their difference is therefore zero, again leaving h_{F^-}. The reconstruction error is therefore $h_{F^-} \in \mathcal{L}_{N-2}$, proving the claim for $k = 2$.

Step ($k > 2$). Assume the claim for $k - 1$, and let P have k facets not parallel to e_1. Two of them have the same nonzero sign. Their intersection is an $(N-3)$-dimensional face C. Let $W := \operatorname{span}\{p - q : p, q \in C\}$, and project orthogonally within H_0 onto $W^\perp$. This is a two-dimensional plane, since $\dim H_0 = N - 1$ and $\dim W = N - 3$. All points of C have the same projection c, because their differences lie in W. There are just two vertices of P outside C, so the projected simplex is a triangle with vertices c and the projections of those two vertices. The chosen facets project to its two sides meeting at c.

The chosen facet normals are orthogonal to W, so their signs against e_1 equal their signs against its projection. This projected direction is nonzero, since the chosen facets are not parallel to e_1. If both sides are upper, moving from c opposite to this direction enters the triangle; if both are lower, moving in this direction enters it. Continuing meets the interior of the opposite edge. The inverse image of the line through c in this direction is a hyperplane in H_0 containing C and parallel to e_1. It cuts the edge joining the two vertices outside C at an interior point and divides P into two $(N-1)$-dimensional simplices. Each keeps one of the two chosen facets and replaces the other by the new facet parallel to e_1. Its other facets retain their normal directions, so each new simplex has exactly $k - 1$ facets not parallel to e_1.

By the induction hypothesis, both new simplices have reconstruction errors in $\mathcal{L}_{N-2}$. Apply (52) to P and to each facet split by the cut. The reconstruction error of P is the sum of these two errors, plus a linear combination of support functions of cutting sections of dimension at most $N - 2$ and $\mathcal{T}$ applied to support functions of their facets of dimension at most $N - 3$. Equation (56) puts every correction in $\mathcal{L}_{N-2}$. This proves the claim for k and completes the induction.

Conclusion for f. Apply (57) to every simplex support-function term in the expansion of f, keeping its original coefficient. A facet F parallel to e_1 has $\langle \nu_F, e_1 \rangle = 0$, so its term in the reconstruction formula is zero and can be omitted. For each remaining occurrence of a facet, choose ν_F outward from its simplex and put

$$u_F := \frac{\nu_F}{\langle \nu_F, e_1 \rangle}, \qquad \langle u_F, e_1 \rangle = 1. \tag{58}$$

Thus the coordinate with index 1 is normalized to 1; the coordinate with index 0 remains zero because $u_F \in H_0$. Oppositely oriented parallel normals give the same u_F. Only the normal is rescaled: the support function and its coefficient are unchanged.

Let $\mathcal{U}$ be the finite set of vectors u_F obtained this way. For each $u \in \mathcal{U}$, define G_u by adding all facet support-function terms with $u_F = u$, retaining their signs and weights. An occurrence of F as a facet of $K_{\pi,S,\omega}$ contributes $(-1)^{N+|S|} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F / N!$, while an occurrence as a facet of Δ contributes $-\operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F$. These are precisely the terms with the same normalized normal, including all repetitions.

If $\langle \nu_F, e_1 \rangle > 0$, the direction u_F exposes F; if it is negative, $-u_F$ exposes F. The opposite face of that simplex is a single vertex. Thus (54) gives the required positive or negative facet term up to a linear function. Directions exposing no facet contribute only support functions of faces of dimension at most $N - 3$, and (37) handles the subdivision remainder. Consequently,

$$G_u - D_u f \in \mathcal{L}_{N-3} \qquad (u \in \mathcal{U}). \tag{59}$$

The assumption (50) puts every G_u in $\mathcal{L}_{N-3}$. By linearity, the sum of the reconstructed facet terms is $\mathcal{T}(\sum_{u \in \mathcal{U}} G_u)$, which lies in $\mathcal{L}_{N-2}$ by (56). Its difference from f consists of the reconstruction and subdivision remainders, also in $\mathcal{L}_{N-2}$. Hence $f \in \mathcal{L}_{N-2}$. This finishes the proof of Step 7.2.

Step 7.3 (Prove the class identity by induction).

Claim (induction on N). The class identity (42) holds for every $N \geq 3$.

Start ($N = 3$). Here $r = 1$, so the nonempty cut set S must be $\{1\}$. The tree has the single edge $\pi(1) \to \pi(2)$, with $q = 1$ and $p(1) = 0$. Equation (38) gives

$$\begin{aligned} x_{\pi(0)} + U_{\pi,\{1\}}(x) &= \max\{x_{\pi(0)}, x_{\pi(2)} + \sigma(x_{\pi(1)} - x_{\pi(2)})\} \\ &= \max\{x_{\pi(0)}, x_{\pi(1)}, x_{\pi(2)}\} = \max_3(x). \end{aligned} \tag{60}$$

Each term has coefficient $(-1)^{3+1}/3! = 1/6$, so the six permutations give

$$F_3(x) = \frac{1}{6} \sum_{\pi \in \mathrm{Sym}\{0,1,2\}} \max_3(x) = \max_3(x). \tag{61}$$

On H_0, $x_0 = 0$ and $\max_3(x) = \max\{0, x_1, x_2\} = h_\Delta(x)$. Hence $f = 0 \in \mathcal{L}_1$, proving (42) for $N = 3$.

Step ($N \geq 4$). Assume the class identity for $N - 1$. By Step 7.2, it suffices to prove $D_u f \in \mathcal{L}_{N-3}$ for every $u \in H_0$. By (35), applying D_u replaces each support function h_P by $h_{P^u} - h_{P^{-u}}$: we select the maximizing and minimizing faces in direction u and subtract their support functions. The linear apex terms disappear. For Δ, these faces contain exactly the vertices with largest and smallest coordinate values u_i, including $u_0 = 0$. For a caterpillar base, the segment $[0, \ell_{ij}]$ remains free when $u_i = u_j$; its maximizing endpoint is ℓ_{ij} when $u_i > u_j$ and 0 when $u_i < u_j$. The minimizing face uses the opposite choices.

We discard support-function contributions already in $\mathcal{L}_{N-3}$ and cancel the remaining ones. Induction is needed only when one of $u_0, \ldots, u_{N-1}$ differs from the common value of all the others: the exposed simplex facet and the surviving caterpillar sum then give the same problem with $N - 1$ inputs.

Discarding contributions in $\mathcal{L}_{N-3}$. Fix $u \neq 0$, since $D_0 f = 0$. A face of $Q_{\pi,S}$ contained in its base has a support function that is a linear function plus support functions of segments, hence belongs to $\mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$; the apex alone contributes zero. If at least two edges join labels with unequal u_i, at most $N-4$ segments remain free, so even adjoining the apex gives dimension at most $N - 3$. Thus we need only faces containing the apex and an exposed base face, with at most one edge joining unequal values.

Removing that edge splits the tree into two components on which the u_i are constant. If it is a spine edge, the tie between apex and base forces the apex label's value to equal one of these two values. If it is a leaf edge, write d for the leaf value and c for the other tree values. The tie forces the apex label's value to be $c + \sigma(c - d)$, either c or $2c - d$. With no such edge, all tree values agree. Consequently, only two distinct coordinate values or the three values $c+t, c-t, c, \ldots, c$ with $t \neq 0$ can contribute. All other directions are already handled by the dimension bound.

Cancellation for three distinct values. For $c + t, c - t, c, \ldots, c$, exchanging the two exceptional simplex vertices and returning to H_0 acts as a linear reflection sending u to $-u$. Permutation symmetry preserves F_N up to a linear function. Comparing the one-sided derivatives at u and $-u$ under this reflection, using positive homogeneity, gives

$$D_u(F_N|_{H_0}) \in \mathcal{L}_0. \tag{62}$$

The two exposed simplex faces are vertices, so their support functions are linear as well.

Cancellation for two values, each repeated. In each remaining caterpillar term, $u_{\pi(1)}, \ldots, u_{\pi(k)}$ equal one value and $u_{\pi(k+1)}, \ldots, u_{\pi(r+1)}$ the other, with $k \in S$. If the apex label has the latter value, toggle membership of $k - 1$ in S. Otherwise, exchange $\pi(k + 1), \pi(k + 2)$ and toggle membership of $k + 1$. Each value occurs at least twice, so these operations are defined. They pair terms with opposite coefficients whose facet differences agree modulo $\mathcal{L}_1$. Each exposed simplex face has at most $N - 2$ vertices and therefore dimension at most $N - 3$. Hence $D_u f \in \mathcal{L}_{N-3}$, without using induction.

One exceptional coordinate: the induction hypothesis. It remains to consider two values, one occurring only once. Permute that label to 0 and subtract its value to return to H_0. This invertible linear change preserves each $\mathcal{L}_k$ and changes f only by a linear term, annihilated by D. Rescaling the direction, and reversing the sign of D_u if necessary, reduces to $u = (0, 1, \dots, 1)$.

This direction exposes the $(N-2)$-dimensional facet $\Delta' := \operatorname{conv}\{e_1, \dots, e_{N-1}\}$ and the opposite vertex 0. Hence

$$D_u h_\Delta(x) = h_{\Delta'}(x) = \max\{x_1, \dots, x_{N-1}\} \qquad (x \in H_0). \tag{63}$$

Here is the exact use of the induction hypothesis. Apply the $(N-1)$-input class identity after the linear substitution $x \mapsto (0, x_2 - x_1, \dots, x_{N-1} - x_1)$. Restoring x_1 changes the error only by a linear function. Since $(N-1) - 2 = N - 3$, this gives

$$\big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big) \in \mathcal{L}_{N-3}. \tag{64}$$

Thus we must match the exposed caterpillar support functions with the lower-arity sum. We prove

$$D_{(0,1,\dots,1)}(F_N|_{H_0}) - \big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1})\big) \in \mathcal{L}_{N-3}. \tag{65}$$

Delete label 0 from each contributing caterpillar. Each resulting caterpillar with a nonempty cut set has $N-2$ pendant insertions and two spine-end insertions that recover its preimages. A pendant insertion reverses both the coefficient sign and the facet-difference sign; an end insertion preserves both. Each contributes $1/N$ of the lower-arity summand modulo $\mathcal{L}_{N-3}$, so the N preimages recover its coefficient in F_{N-1}.

The extra terms are those whose deletion leaves an empty cut set, hence a rooted star. Each endpoint-insertion family reduces to its lower-arity support function modulo $\mathcal{L}_1$, with factor $1/N$. Fix its root and let $d_1, \dots, d_q$, where $q = N - 2$, be its coordinate minus each other remaining coordinate. Put

$$T_A := \sigma\left(\sum_{i \in A} d_i + \sum_{i \notin A} \sigma(d_i)\right). \tag{66}$$

The Boolean alternating identity gives

$$\sum_A (-1)^{|A|} T_A = 0 \quad \Longrightarrow \quad \sum_i T_{\{i\}} = T_\varnothing + \sum_{|A| \geq 2} (-1)^{|A|} T_A. \tag{67}$$

The empty term is a sum of support functions of segments. For $|A| \geq 2$, the polytope with support function T_A has one head direction and $q - |A|$ segment directions, hence dimension at most $1 + q - |A| \leq q - 1 = N - 3$. Thus the singleton sum lies in $\mathcal{L}_{N-3}$. Averaging over the possible apex labels puts the extra star sum there as well, proving (65).

Combining (63), (64), and (65) now gives

$$\begin{aligned} D_u f(x) = {} & \big[D_u(F_N|_{H_0})(x) - F_{N-1}(x_1, \dots, x_{N-1})\big] \\ & + \big[F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big]. \end{aligned} \tag{68}$$

The first bracket belongs to $\mathcal{L}_{N-3}$ by coordinate deletion, the second by the induction hypothesis. This proves the bound for every direction. Step 7.2 gives $f \in \mathcal{L}_{N-2}$, completing the induction, and Step 7.1 gives the dictionary representation. This finishes the proof of Theorem 7. □

4 NETWORK SIZE

We prove Theorem 2 directly. First we bound the number of distinct caterpillar face functions. Then we construct one shared first layer and account for both widths.

Proof of Theorem 2. Fix $N \geq 3$. Theorem 7 and Lemma 6 give the exact canonical expansion of $\max_N$.

Step 2.1 (Count the caterpillar face functions). Every face function is again an all-hinge unit: collecting the linear states into the head,

$$U^\tau_{h,L} = U_{h',M}, \qquad h' := h + \sum_{\tau(\ell) = \ell} \ell, \qquad M := \{\ell \in L : \tau(\ell) = \sigma \circ \ell\}. \tag{69}$$

We call such a pair (h, M), a linear head and a finite set of edge functions, a *description* of $U_{h,M}$. Distinct descriptions may represent the same function, so the number of descriptions bounds the number of functions. By (10), applied to each $\ell_{ij} \in M$ with $i > j$, the function $U_{h,M}$ also has the *normalized description*

$$\begin{aligned} M^+ &:= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in M\}, \\ h^+ &:= h + \sum_{\ell_{ij}\in M,\ i>j} \ell_{ij}, \end{aligned} \tag{70}$$

in which every hinge is oriented as $x_i - x_j$ with $i < j$. No edge function of a caterpillar is the negative of another one, so M and M^+ have the same size.

For a caterpillar description (π, S), the spine endpoints are labelled $\pi(1)$ and $\pi(q+1)$, and every label other than $\pi(1)$ and the apex is the child of exactly one edge. Hence the normalized description of $U_{\pi,S}$ is

$$\begin{aligned} h^+_{\pi,S}(x) &= \sum_{\ell_{ij}\in L_{\pi,S}} x_{\max(i,j)} + x_{\pi(1)} + x_{\pi(q+1)} - \sum_{i=0}^{N-1} x_i, \\ L^+_{\pi,S} &= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in L_{\pi,S}\}. \end{aligned} \tag{71}$$

It depends only on the undirected tree, the apex, and the unordered pair of spine endpoints, not on the order of the leaves at a spine vertex or on the orientation of the spine. For a spine with m vertices, choose its vertex set in $\binom{N}{m}$ ways, its order up to reversal in $m!/2$ ways, its apex in $N - m$ ways, and the attachment of each of its $N - m - 1$ leaves in m ways. Reversal has no fixed point since $m \geq 2$. Every caterpillar has such data, so by (6)

$$\left|\{(h^+_{\pi,S}, L^+_{\pi,S}) : (\pi, S) \text{ a caterpillar description}\}\right| \leq C_N. \tag{72}$$

Reversing an edge function swaps its omitted and linear states, so normalizing before or after choosing the states gives the same descriptions. Every face function $U^\tau_{\pi,S}$ therefore has a normalized description obtained from $(h^+_{\pi,S}, L^+_{\pi,S})$ by choosing one of three states for each of its $N - 2$ edge functions. With (72), there are at most $3^{N-2}C_N$ distinct caterpillar face functions.

Among them are the $N(N-1)$ pair hinges $\sigma \circ \ell_{ij}$ with $i \neq j$: for the all-cut path $S = \{1, \ldots, r\}$ with apex $\pi(0) = j$ and last spine vertex $\pi(r+1) = i$, omitting every edge function gives

$$U^0_{\pi,S} = \sigma \circ \ell_{ij} \qquad (0 \leq i \neq j < N). \tag{73}$$

By (10) these lie in the span of the auxiliary functions and need no second-layer unit. Let K be the set of the remaining caterpillar face functions with nonzero coefficient in (27). Then

$$|K| \leq 3^{N-2}C_N - N(N-1). \tag{74}$$

This finishes the proof of Step 2.1.

Step 2.2 (Compile the network and establish its widths). Let $H := \sum_{g\in K} c_g g$ and

$$R := \max\nolimits_N - H. \tag{75}$$

By (27), R is a linear combination of the auxiliary functions (20) and of pair hinges, hence of the auxiliary functions alone by (10). Use the shared first-layer features

$$\sigma(x_i - x_j)\ (i < j), \qquad \sigma(x_i - x_0)\ (i > 0), \qquad \sigma(x_0),\ \sigma(-x_0). \tag{76}$$

Their number is

$$\binom{N}{2} + (N-1) + 2 = \binom{N}{2} + N + 1 = h_1. \tag{77}$$

By (10), these features recover all coordinates,

$$\begin{aligned} x_0 &= \sigma(x_0) - \sigma(-x_0), \\ x_i &= x_0 + \sigma(x_i - x_0) - \sigma(x_0 - x_i) \qquad (i > 0), \end{aligned} \tag{78}$$

and then, by (10) once more, every reversed pair hinge $\sigma(x_j - x_i)$ with $i < j$. Thus every linear function and every pair hinge in either orientation is a linear combination of (76). In particular, the outer preactivation of each unit $g \in K$, a linear function plus pair hinges, is such a combination.

To recover R as well, note that H is translation invariant by (21), while $\max_N(x) = x_0 + \max_N(x - x_0\mathbf{1})$. Hence

$$R(x) = x_0 + R(x - x_0\mathbf{1}). \tag{79}$$

In the auxiliary expansion of $R(x - x_0\mathbf{1})$, the coordinate features become $\sigma(\pm(x_i - x_0))$, which vanish for $i = 0$ and are features or reversed features for $i > 0$, while pair hinges are unchanged. With (78) for the added x_0, the function R is a linear combination of (76).

Use one second-layer neuron for each $g \in K$ and two more for

$$R = \sigma(R) - \sigma(-R). \tag{80}$$

The affine output sums the retained terms and this difference, giving $H + R = \max_N$. By (74), the second-layer width is at most

$$|K| + 2 \le 3^{N-2}C_N - N(N-1) + 2 = h_2. \tag{81}$$

Zero padding realizes the stated widths exactly.

For the growth rates,

$$\binom{N}{m}\frac{m!}{2} \le N^N, \tag{82}$$

and each remaining factor in (6) is at most N^N. There are at most N terms, giving, for $N \ge 3$,

$$C_N \le N^{4N}, \tag{83}$$

$$h_2 \le 2N^{5N}, \tag{84}$$

$$h_1 \le 4N^2. \tag{85}$$

Taking logarithms gives (8). This finishes the proof of Theorem 2. □

Formalization (`Lean_10`) Linked definitions, lemmas, and theorems are audited against exact types; only `propext`, `Classical.choice`, and `Quot.sound` are allowed. The repository contains the maximum construction and its proof. The general CPWL results in the introduction are cited background results; the equation links in the Wang–Sun theorem lead to its published source. Claude and Codex assisted with formalization and manuscript preparation.

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